MEASUREMENT & INSTRUMENTATION LABORATORY

Automated Temperature Data Acquisition and Control System

A LabVIEW-Based Virtual Instrumentation Approach to Real-Time Threshold Monitoring, Automated Actuation, and Continuous Data Logging

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SECTION 01

Problem Statement

- Manual/analog temperature monitoring in laboratory and industrial environments is discontinuous, operator-dependent, and prone to human error.
- Absence of instant, threshold-triggered alerts increases the risk of equipment damage from overheating or under-cooling conditions.
- Conventional thermometry offers no built-in mechanism for continuous data logging, hindering trend analysis and audit traceability.
- Sensing, decision-making, and actuation are typically handled as disjointed subsystems, increasing overall response latency.
- Constant human supervision of critical thermal processes raises operational cost and introduces reaction delay.
- No low-cost virtual-instrumentation framework exists to validate control logic prior to physical hardware deployment.

KEY RISK FACTORS

Delayed Response

No real-time threshold alerts

Data Loss

No continuous data logging

Thermal Damage

Undetected threshold breach

SECTION 02

Objective

- Design and implement a LabVIEW-based virtual instrumentation system for continuous, simulated temperature acquisition.
- Implement Boolean comparator logic to classify the sensed temperature into High, Low, and Moderate zones in real time.
- Automate actuator response — Cooling Fan and Heater indicators — directly from threshold-crossing events, without manual intervention.
- Provide real-time graphical visualization of the temperature trend using a Waveform Chart.
- Establish persistent, timestamped data logging to an Excel-compatible file via the Write To Measurement File Express VI.
- Validate the reliability, repeatability, and timing consistency of the acquisition-control cycle under continuous While Loop execution.

DESIGN TARGETS

3 Temperature Zones

High / Moderate / Low classification

500 ms Sampling

Fixed loop iteration rate

Live Logging

Timestamped Excel output

SECTION 03

Theory

- Virtual Instrumentation (VI) replaces conventional hardware measurement instruments with software-defined, reconfigurable equivalents.
- LabVIEW follows a Data Flow Programming model — a node executes only after all of its required inputs become available.
- Comparator functions (Greater Than / Less Than) evaluate the sensor signal against fixed thresholds and output a Boolean result.
- Boolean logic (AND) combines two comparator outputs to isolate a bounded "Moderate Temperature" band between the high and low limits.
- A While Loop with a fixed iteration delay (500 ms, via the Wait function) governs the sampling rate, emulating a real-time acquisition cycle.
- A Waveform Chart retains a rolling history of plotted values, unlike a Waveform Graph which redraws only the current data set.

CORE CONCEPTS

Dataflow Execution

A node fires once inputs are ready

Boolean Comparators

Threshold-based decision logic

Iterative Sampling

While Loop with Wait (ms)

SECTION 04

Methodology

- A Simulated Temperature Sensor sub-VI generates continuous DBL-type data representing an RTD/thermocouple-equivalent analog output.
- The sensor signal is wired in parallel to four comparator blocks: High (> 35°C), Low (< 14°C), and a bounded pair (15 / 34) for the Moderate band.
- Boolean outputs from the comparators directly drive the HIGH TEMP, LOW TEMP, and MODERATE TEMP LED indicators on the front panel.
- The HIGH TEMP and LOW TEMP conditions additionally actuate the COOLING FAN and HEATER Boolean control indicators respectively.
- The live sensor signal is simultaneously routed to a Waveform Chart for real-time trend visualization and to the Write To Measurement File Express VI for logging.
- A front-panel STOP button and a 500 ms Wait function together govern loop termination and the fixed sampling interval.

PROCESS FLOW

Sense

Simulated sensor signal (DBL)

Compare & Actuate

Thresholds drive fan/heater LEDs

Log & Display

Waveform Chart + Excel file

SECTION 05

Block Diagram

- While Loop forms the core execution structure, iterating continuously until the STOP condition is triggered.
- Wait (ms) function fixed at 500 ms sets the loop's iteration rate, simulating a real-time sampling interval.
- Simulated Temperature Sensor output branches into four parallel comparator paths for independent threshold checks.
- Greater Than (35) and Less Than (14) blocks drive the HIGH TEMP and LOW TEMP indicators directly.
- Comparators at 15 and 34 feed an AND gate to isolate the Moderate Temperature band.
- Write To Measurement File and the Waveform Chart both tap the live sensor signal for parallel logging and display.

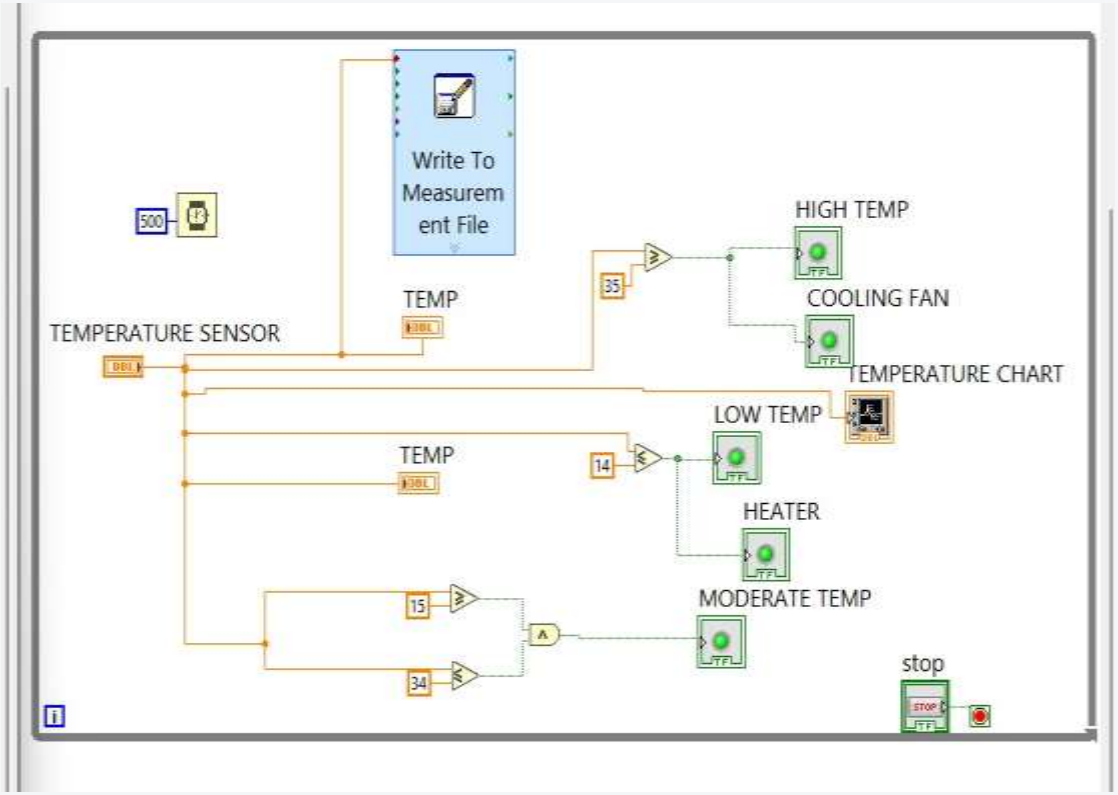


Fig. 1 — LabVIEW Block Diagram (control logic)

SECTION 06

Practical Circuit Diagram

- The Front Panel is the virtual equivalent of a practical instrumentation circuit, replacing physical meters, relays, and switches.
- TEMPERATURE SENSOR uses a horizontal fill-slide indicator (-20°C to 50°C) to emulate an analog sensor readout stage.
- The DISPLAY UNIT pairs a colour-coded analog dial (green-yellow-red band) with a digital TEMP readout for redundant indication.
- HIGH TEMP, MODERATE TEMP, and LOW TEMP LEDs emulate relay-driven indicator lamps in a hardware control circuit.
- COOLING FAN and HEATER LEDs represent the actuator outputs that would drive relay-switched fan/heater loads in deployment.
- The TEMPERATURE CHART plots amplitude versus time — the virtual equivalent of a physical chart recorder/data logger.

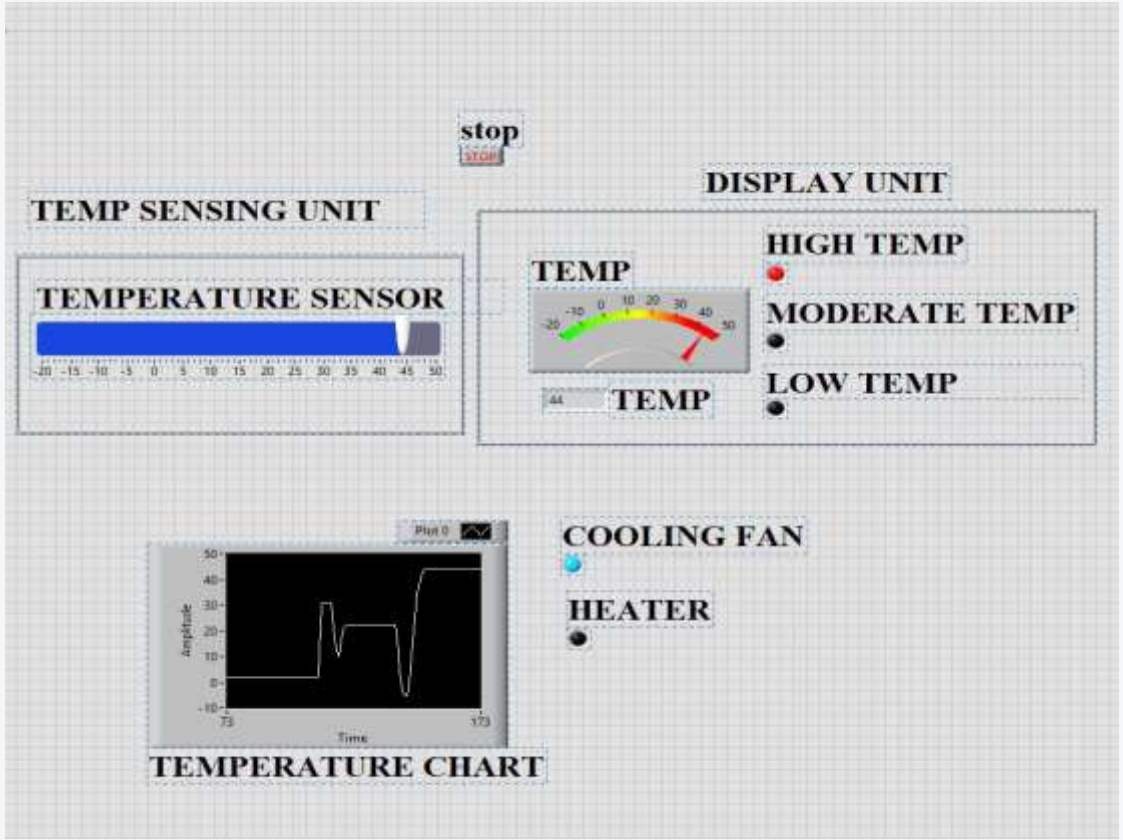
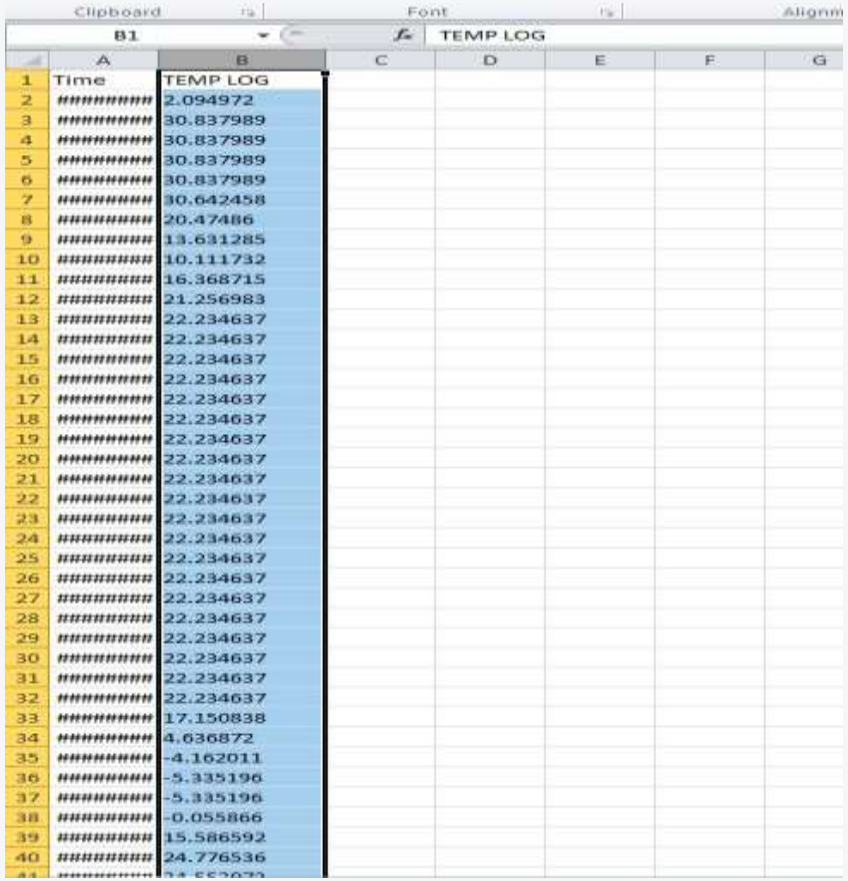


Fig. 2 — Virtual Front Panel (practical equivalent circuit)

SECTION 07

Outcomes & Results

- The system acquired and displayed simulated temperature values continuously, without interruption, over the full test run.
- Comparator logic correctly asserted HIGH TEMP / COOLING FAN above 35°C and LOW TEMP / HEATER below 14°C.
- The MODERATE TEMPERATURE indicator activated only within the 15-34°C band, confirming correct AND-gate operation.
- The Waveform Chart captured clear transitions between zones, including dips near -5°C and peaks approaching 45°C.
- The Write To Measurement File VI generated a clean, timestamped Excel log (Time, TEMP LOG columns) for offline analysis.
- Logged entries confirmed consistent ~500 ms sampling intervals, validating the repeatability of the acquisition-control cycle.



	A	B
1	Time	TEMP LOG
2	##########	2.094972
3	##########	30.837989
4	##########	30.837989
5	##########	30.837989
6	##########	30.837989
7	##########	30.642458
8	##########	20.47486
9	##########	13.631285
10	##########	10.111732
11	##########	16.368715
12	##########	21.256983
13	##########	22.234637
14	##########	22.234637
15	##########	22.234637
16	##########	22.234637
17	##########	22.234637
18	##########	22.234637
19	##########	22.234637
20	##########	22.234637
21	##########	22.234637
22	##########	22.234637
23	##########	22.234637
24	##########	22.234637
25	##########	22.234637
26	##########	22.234637
27	##########	22.234637
28	##########	22.234637
29	##########	22.234637
30	##########	22.234637
31	##########	22.234637
32	##########	22.234637
33	##########	17.150838
34	##########	4.636872
35	##########	-4.162011
36	##########	-5.335196
37	##########	-5.335196
38	##########	-0.055866
39	##########	15.586592
40	##########	24.776536

Fig. 3 — Logged Excel Output (Time, TEMP LOG)

SECTION 08

Advantages

- Eliminates the need for physical prototype hardware during initial design and validation, reducing cost and development time.
- Real-time visual feedback (LEDs, dial gauge, chart) improves operator situational awareness and shortens response latency.
- Automated, timestamped data logging removes manual record-keeping errors and enables long-term trend analysis.
- Threshold values, sampling rate, and control logic can be reconfigured entirely in software, without any hardware rewiring.
- Modular VI-based architecture allows straightforward scaling to additional sensors, actuators, or control loops.
- Graphical (dataflow) programming improves code readability and simplifies debugging relative to text-based control code.

AT A GLANCE

Zero Hardware Cost

Fully simulated in software

Instant Response

Real-time actuation logic

Full Traceability

Automatic timestamped logging

SECTION 09

Applications

- Industrial process control — furnace, boiler, and HVAC temperature regulation systems.
- Environmental and greenhouse monitoring for climate-sensitive agricultural production.
- Server room and data-centre thermal management with automated cooling response.
- Cold-chain logistics and food storage compliance monitoring against regulatory limits.
- Laboratory instrumentation for calibration, testing, and general-purpose research data acquisition.
- Educational and training platforms for teaching virtual instrumentation and control-system fundamentals.

WHERE IT FITS

Industrial Control

Furnaces, HVAC, boilers

Environmental

Greenhouses, cold-chain logistics

Research & Training

Laboratories and education

SECTION 10

Conclusion

- The project successfully demonstrates an integrated data acquisition and control system built on LabVIEW virtual instrumentation.
- Boolean comparator-based logic reliably classified temperature states and triggered the appropriate actuator response in each case.
- Real-time visualization together with persistent data logging provides a complete monitoring and record-keeping solution.
- The design validates the feasibility of migrating from a simulated sensor to a physical hardware sensor/actuator interface.
- The While Loop-driven architecture confirms suitability for continuous, unattended industrial monitoring applications.
- The system forms a strong foundation for future enhancement with PID control and IoT-based remote monitoring.

SECTION 11

References

- National Instruments, "LabVIEW User Manual," NI Corporation, Austin, TX.
- National Instruments, "Write To Measurement File Express VI" — LabVIEW Help Documentation, NI Corporation.
- J. Travis and J. Kring, LabVIEW for Everyone: Graphical Programming Made Easy and Fun, 3rd ed., Prentice Hall, 2006.
- R. Bitter, T. Mohiuddin, and M. Nawrocki, LabVIEW: Advanced Programming Techniques, 2nd ed., CRC Press, 2007.
- Measurement and Instrumentation Laboratory Manual, Department of Electrical and Electronics Engineering, VIT.
- IEEE Instrumentation and Measurement Society, "Guide for Temperature Measurement and Instrumentation Systems."

CITATION NOTES

IEEE Style

Consistent citation format used

6 Sources Cited

Manuals, texts, and standards

NI Documentation

Primary technical reference

THANK YOU

Measurement and instrumentation Project